PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

LAB 7: Fuzzing for Input Validation Bugs

Course: Software Engineering Lab | Fuzzing Library: Hypothesis

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Repository: https://github.com/rajatkooge5-learn/L6_L7_50_recycling_rewards

1. Objective & Overview

The objective of this lab is to explore fuzz testing as a dynamic security and quality assurance technique. Using the Hypothesis library, we generate random, unexpected, and malformed inputs to uncover edge cases, unhandled exceptions, and input validation bugs in recycling_rewards / processor.py. We then apply defensive programming to harden the application against crashes.

2. Step 2: Run Buggy Application (processor.py)

Command: python processor.py
The application was executed with expected, valid user inputs. In standard operation, the program functions without issue. However, it lacks all input validation and boundary checks.

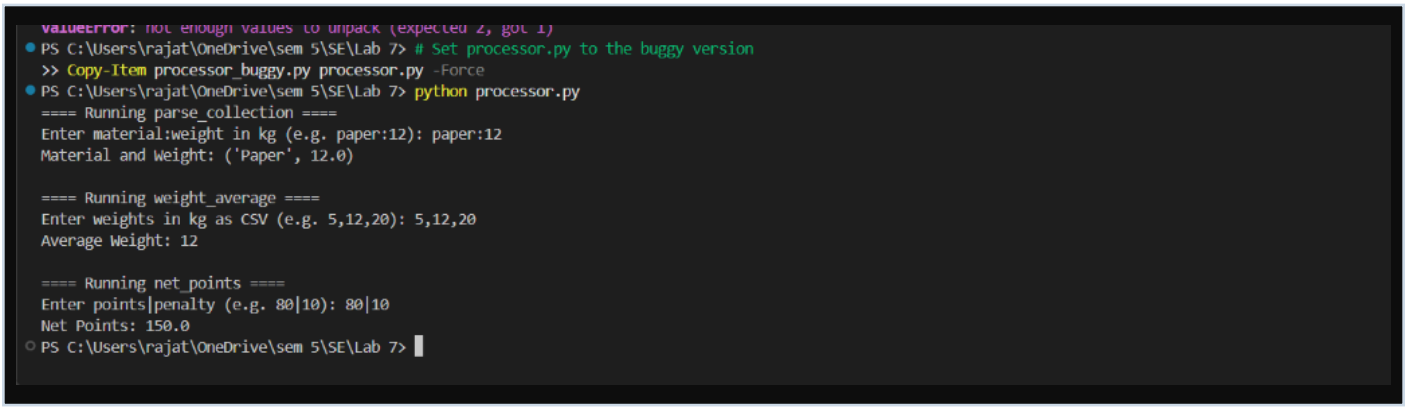


Figure 1: Step 2 - Buggy application executing with standard valid inputs.

3. Step 3 & Step 4: Hypothesis Fuzz Testing & Failure Discovery

In Step 3, property-based tests were authored in test_processor.py using Hypothesis (@given decorator) with st.text() | st.none() strategies. In Step 4, tests were run using: python -m pytest test_processor.py -v Hypothesis generated hundreds of edge-case inputs and immediately broke all three functions:

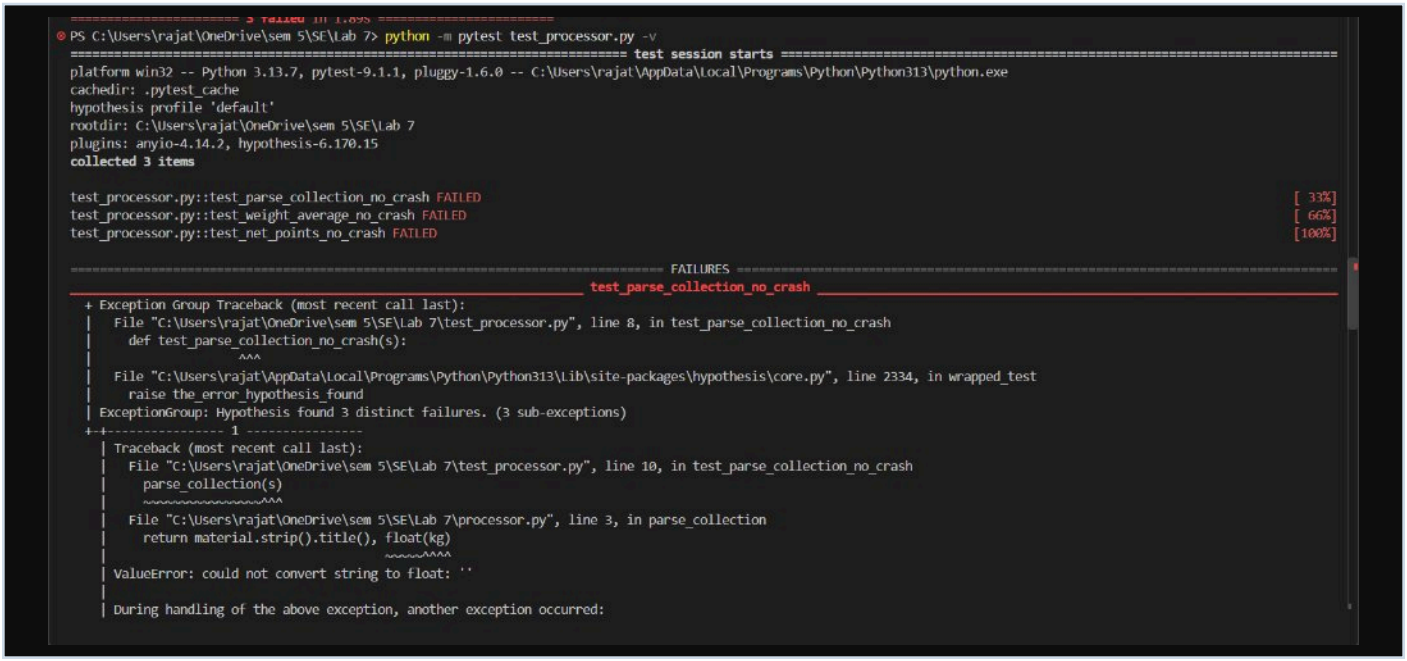


Figure 2: Step 4 - Hypothesis uncovering crashes and unhandled exceptions across all 3 functions.

4. Edge Cases & Bugs Discovered by Hypothesis

Function	Failing Input (Minimal)	Exception Raised	Root Cause
parse_collection	s = '' or s = 'abc'	ValueError: not enough values	No delimiter ":" present
parse_collection	s = ':'	ValueError: could not convert	Empty weight substring
parse_collection	s = None	AttributeError: 'NoneType' has	Missing NoneType validation
weight_average	s = '' or s = ' '	ValueError: invalid literal fo	Empty token after split
weight_average	s = None	AttributeError: 'NoneType' has	Missing NoneType validation
weight_average	s = ',,'	ZeroDivisionError: integer div	Empty values list (len == 0)
net_points	s = '' or s = '80'	ValueError: not enough values	No delimiter "|" present
net_points	s = '|'	ValueError: could not convert	Empty float tokens
net_points	s = None	AttributeError: 'NoneType' has	Missing NoneType validation

5. Step 5: Fixing the Application (Defensive Programming)

All three functions in processor.py were hardened with comprehensive defensive validation:
1. Type checking: isinstance(text, str) ensures NoneType or non-string inputs are safely handled.
2. Boundary & delimiter checks: Verifying len(parts) == 2 before unpacking.
3. Numeric conversion safety: Enclosed in try-except (ValueError, TypeError) blocks.
4. Special float checks: math.isnan() and math.isinf() checks prevent NaN/Infinity poisoning.
5. Graceful fallbacks: Returning None (parse_collection), 0 (weight_average), or 0.0 (net_points).

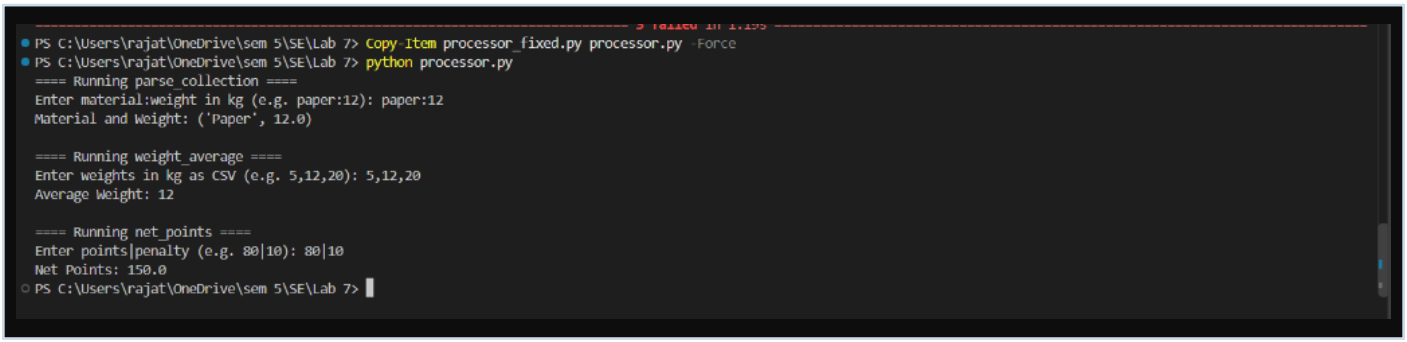


Figure 3: Step 5 - Fixed application executing cleanly with valid inputs.

6. Step 6: Re-Running Fuzz Tests on Fixed Code

Command: python -m pytest test_processor.py -v
The fuzz tests were executed against the fixed processor.py. Hypothesis subjected each function to hundreds of randomized unicode strings and None values. All property tests passed with 0 failures.

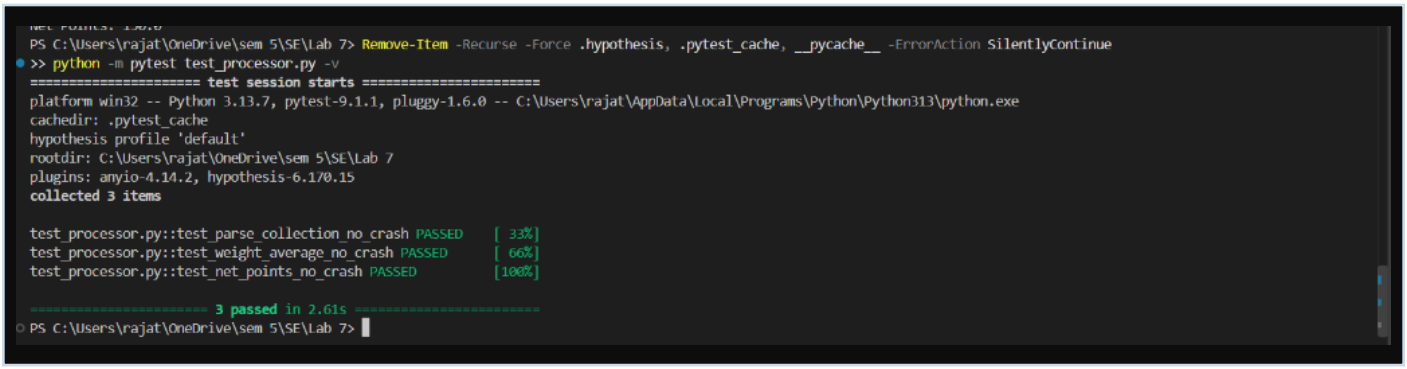


Figure 4: Step 6 - All Hypothesis fuzz tests passing cleanly (100% pass rate).

7. Deliverables & Submission Verification

? Deliverable 1: PDF Document with Screenshots (Lab7_PES1UG24CS362_Report.pdf)
? Deliverable 2: Fixed application python script (processor.py)
? Deliverable 3: Updated test_processor.py code using Hypothesis fuzzing

Repository Verification:
All deliverables, code files, and screenshots are synchronized to the assigned GitHub repository:
https://github.com/rajatkooge5-learn/L6_L7_50_recycling_rewards
Folder: Lab 7/